EXPERIMENT NO: 06

Experiment Title: Use JavaScript string functions and regular expressions (regex) for email validation, data extraction, and text analysis.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Theory:

JavaScript String Methods: Strings in JavaScript represent sequence data. Built-in methods allow searching, extracting, transforming, and formatting textual data effortlessly.

- **Search & Inspection:** indexOf(), includes(), startsWith(), endsWith().
- **Extraction:** slice(), substring(), split().
- **Transformation:** toLowerCase(), toUpperCase(), trim(), replace().

Regular Expressions (Regex): A regular expression is a pattern used to match character combinations in strings. In JavaScript, regex patterns are enclosed between slashes /pattern/flags or defined using the RegExp object.

- **Email Pattern Matching:** Uses pattern checks like [a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/ to confirm valid email formats (username@domain.extension).
- **Regex Methods:**
  - test(): Evaluates a string against a pattern and returns true or false.
  - match(): Extracts matching substrings into an array.
  - replace(): Finds pattern matches and substitutes them with a replacement string.

Experiment Program Code

Code 1: String_Fn.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>JavaScript String Functions</title>
</head>

<body>

    <input type="text" id="input" placeholder="Enter input">

    <div id="output"></div>

    <script>

        let input = document.getElementById("input");
        let output = document.getElementById("output");
```

```
        input.addEventListener("input", function () {

            let str = input.value;

            // 1. String()
            let result1 = String(str);

            // 2. match()
            let result2 = str.match(/Hello/g);

            // 3. replace()
            let result3 = str.replace("World", "JavaScript");

            // 4. indexOf()
            let result4 = str.indexOf("World");

            // 5. trim()
            let result5 = str.trim();

            // 6. includes()
            let result6 = str.includes("World");

            // 7. lastIndexOf()
            let result7 = str.lastIndexOf("Hello");

            // 8. toUpperCase()
            let result8 = str.toUpperCase();

            // 9. toLowerCase()
            let result9 = str.toLowerCase();

            // 10. split()
            let result10 = str.split(" ");

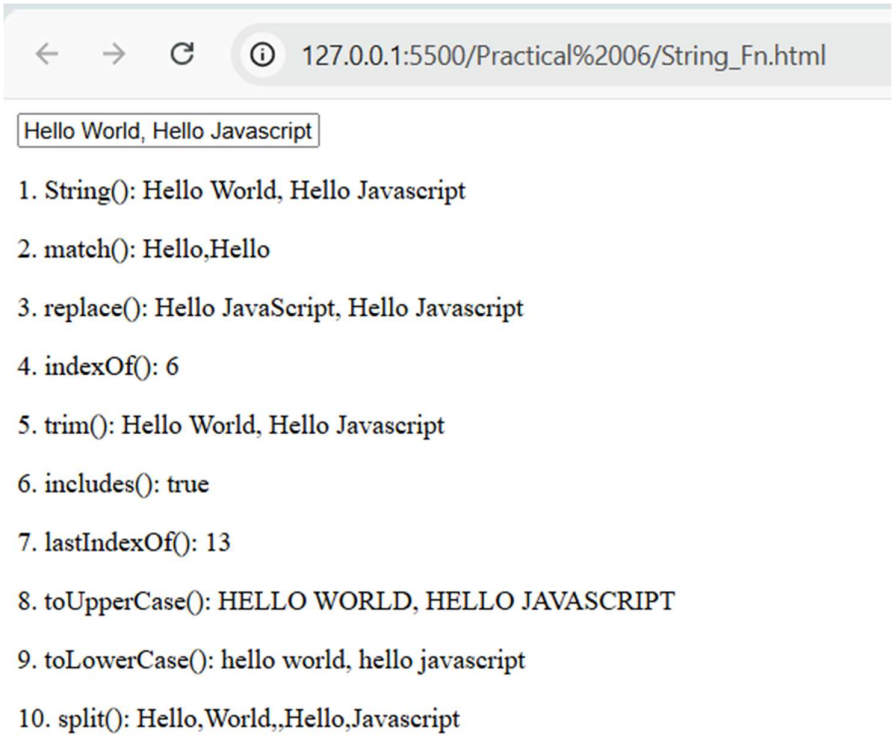
            // Display results sequentially
            output.innerHTML = `
                <p>1. String(): ${result1}</p>
                <p>2. match(): ${result2}</p>
                <p>3. replace(): ${result3}</p>
                <p>4. indexOf(): ${result4}</p>
                <p>5. trim(): ${result5}</p>
                <p>6. includes(): ${result6}</p>
                <p>7. lastIndexOf(): ${result7}</p>
                <p>8. toUpperCase(): ${result8}</p>
                <p>9. toLowerCase(): ${result9}</p>
                <p>10. split(): ${result10}</p>
            `;

        });

    </script>

</body>
</html>
```

4. Output



5. Case Study Title

Design a Student Portfolio Setup Application using Javascript functions and regular expressions

6. Case Study Program Code

Casestudy_06_Shweta_Kushwaha_1001.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Portfolio Setup</title>

    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            font-family: Arial, sans-serif;
        }

        body {
            min-height: 100vh;
            background: #f7f1e8;
            color: #3d342c;
            padding: 30px;
```

```
        }

        .container {
            max-width: 1200px;
            margin: auto;
        }

        /* HEADER */

        header {
            text-align: center;
            margin-bottom: 35px;
        }

        header h1 {
            font-size: 38px;
            color: #3d342c;
            margin-bottom: 10px;
        }

        header p {
            color: #81766b;
            font-size: 16px;
        }

        /* MAIN GRID */

        .main-grid {
            display: grid;
            grid-template-columns: 1fr 1fr;
            gap: 24px;
        }

        /* CARDS */

        .card {
            background: #ffffff;
            border: 1px solid #e8ded1;
            border-radius: 20px;
            padding: 26px;
            box-shadow: 0 8px 25px rgba(91, 72, 51, 0.08);
        }

        .card h2 {
            margin-bottom: 20px;
            font-size: 21px;
            color: #4a3d32;
        }

        /* INPUTS */

        .input-group {
            margin-bottom: 16px;
        }

        label {
            display: block;
            margin-bottom: 7px;
            color: #6e6257;
            font-size: 14px;
            font-weight: 600;
        }

        input,
        textarea {
            width: 100%;
            padding: 12px 14px;
            border-radius: 10px;
            border: 1px solid #ded2c4;
            background: #fcfaf7;
            color: #3d342c;
            outline: none;
            font-size: 14px;
            transition: 0.2s;
        }

        input::placeholder,
```

```
        textarea::placeholder {
            color: #aaa096;
        }

        input:focus,
        textarea:focus {
            border-color: #b98562;
            background: #ffffff;
            box-shadow: 0 0 0 3px rgba(185, 133, 98, 0.12);
        }

        textarea {
            height: 100px;
            resize: vertical;
        }

        /* BUTTON */

        .btn {
            width: 100%;
            padding: 13px;
            border: none;
            border-radius: 10px;
            background: #9b6b4d;
            color: white;
            font-size: 16px;
            font-weight: bold;
            cursor: pointer;
            margin-top: 5px;
            transition: 0.2s;
        }

        .btn:hover {
            background: #82583f;
            transform: translateY(-1px);
            box-shadow: 0 5px 15px rgba(130, 88, 63, 0.18);
        }

        /* PORTFOLIO */

        .portfolio {
            text-align: center;
            display: flex;
            flex-direction: column;
            justify-content: center;
        }

        .avatar {
            width: 90px;
            height: 90px;
            margin: 5px auto 15px;
            border-radius: 50%;
            background: #efe1d2;
            color: #82583f;
            display: flex;
            align-items: center;
            justify-content: center;
            font-size: 35px;
            font-weight: bold;
            border: 5px solid #f8f1e9;
        }

        .portfolio h3 {
            font-size: 27px;
            color: #3d342c;
            margin-bottom: 5px;
        }

        .portfolio .email {
            color: #9b6b4d;
            margin-bottom: 18px;
        }

        .bio {
            color: #756a60;
            line-height: 1.6;
            margin-bottom: 20px;
```

```
        }

        /* SKILLS */

        .skills {
            display: flex;
            justify-content: center;
            flex-wrap: wrap;
            gap: 8px;
            margin-bottom: 20px;
        }

        .skill {
            background: #f5ebe0;
            color: #82583f;
            border: 1px solid #e5d2c0;
            padding: 6px 12px;
            border-radius: 20px;
            font-size: 13px;
            font-weight: 600;
        }

        /* LINKS */

        .links {
            display: flex;
            gap: 10px;
            justify-content: center;
        }

        .links a {
            text-decoration: none;
            color: #6d503c;
            background: #f4eee7;
            border: 1px solid #e2d7cb;
            padding: 8px 15px;
            border-radius: 8px;
            font-size: 13px;
            transition: 0.2s;
        }

        .links a:hover {
            background: #eadbcb;
        }

        /* ANALYTICS */

        .analytics {
            grid-column: 1 / -1;
        }

        .stats {
            display: grid;
            grid-template-columns: repeat(4, 1fr);
            gap: 15px;
        }

        .stat {
            background: #fcfaf7;
            border: 1px solid #e6dbcf;
            border-radius: 14px;
            padding: 18px;
            text-align: center;
        }

        .stat .number {
            display: block;
            font-size: 28px;
            font-weight: bold;
            color: #9b6b4d;
            margin-bottom: 5px;
        }

        .stat small {
            color: #81766b;
        }
```

```
        /* VALIDATION */

        .validation {
            margin-top: 25px;
        }

        .validation-item {
            padding: 12px 15px;
            border-radius: 10px;
            margin-bottom: 8px;
            background: #fcfaf7;
            border: 1px solid #e7ddd2;
            display: flex;
            justify-content: space-between;
        }

        .valid {
            color: #5d8766;
        }

        .invalid {
            color: #b65f52;
        }

        /* KEYWORDS */

        .keywords {
            margin-top: 22px;
        }

        .keyword-list {
            margin-top: 10px;
            color: #756a60;
            background: #f8f2eb;
            border-radius: 10px;
            padding: 12px;
        }

        /* FOOTER */

        .footer {
            text-align: center;
            color: #9a9086;
            margin-top: 30px;
            font-size: 13px;
        }

        /* RESPONSIVE */

        @media (max-width: 800px) {

            body {
                padding: 18px;
            }

            .main-grid {
                grid-template-columns: 1fr;
            }

            .analytics {
                grid-column: auto;
            }

            .stats {
                grid-template-columns: repeat(2, 1fr);
            }

            header h1 {
                font-size: 30px;
            }
        }
    </style>
</head>

<body>

<div class="container">
```

```
<header>
    <h1>🎓 Student Portfolio Setup</h1>
    <p>JavaScript String Functions & Regular Expression Validator</p>
</header>

<div class="main-grid">

    <!-- STUDENT INFORMATION -->

    <div class="card">

        <h2>📝 Student Information</h2>

        <div class="input-group">
            <label>Full Name</label>
            <input type="text" id="name"
                   placeholder="Enter your full name">
        </div>

        <div class="input-group">
            <label>Email Address</label>
            <input type="text" id="email"
                   placeholder="example@gmail.com">
        </div>

        <div class="input-group">
            <label>Phone Number</label>
            <input type="text" id="phone"
                   placeholder="9876543210">
        </div>

        <div class="input-group">
            <label>Skills</label>
            <input type="text" id="skills"
                   placeholder="JavaScript, Python, AI, SQL">
        </div>

        <div class="input-group">
            <label>Bio</label>
            <textarea id="bio"
                placeholder="Write something about yourself..."></textarea>
        </div>

        <div class="input-group">
            <label>LinkedIn URL</label>
            <input type="text" id="linkedin"
                placeholder="https://linkedin.com/in/username">
        </div>

        <div class="input-group">
            <label>GitHub URL</label>
            <input type="text" id="github"
                placeholder="https://github.com/username">
        </div>

        <button class="btn" onclick="generatePortfolio()">
            Generate Portfolio
        </button>

    </div>

    <!-- PORTFOLIO PREVIEW -->

    <div class="card portfolio">

        <h2>✨ Portfolio Preview</h2>

        <div class="avatar" id="avatar">S</div>

        <h3 id="previewName">
            Student Name
        </h3>

        <p class="email" id="previewEmail">
            student@example.com
        </p>
```

```
        <p class="bio" id="previewBio">
            Your portfolio bio will appear here.
        </p>

        <div class="skills" id="previewSkills">
            <span class="skill">
                Your Skills
            </span>
        </div>

        <div class="links">

            <a id="linkedinLink"
               href="#"
               target="_blank">
                LinkedIn
            </a>

            <a id="githubLink"
               href="#"
               target="_blank">
                GitHub
            </a>

        </div>

    </div>

    <!-- ANALYTICS -->

    <div class="card analytics">

        <h2>📊 Text Analytics</h2>

        <div class="stats">

            <div class="stat">
                <span class="number" id="charCount">0</span>
                <small>Characters</small>
            </div>

            <div class="stat">
                <span class="number" id="wordCount">0</span>
                <small>Words</small>
            </div>

            <div class="stat">
                <span class="number" id="skillCount">0</span>
                <small>Skills</small>
            </div>

            <div class="stat">
                <span class="number" id="vowelCount">0</span>
                <small>Vowels</small>
            </div>

        </div>

        <!-- REGEX -->

        <div class="validation">

            <h2>🔍 Regex Validation</h2>

            <div class="validation-item">
                <span>Email</span>
                <strong id="emailStatus">
                    Not Checked
                </strong>
            </div>

            <div class="validation-item">
                <span>Phone</span>
                <strong id="phoneStatus">
                    Not Checked
                </strong>
```

```
                </div>

                <div class="validation-item">
                    <span>Name</span>
                    <strong id="nameStatus">
                        Not Checked
                    </strong>
                </div>

                <div class="validation-item">
                    <span>LinkedIn</span>
                    <strong id="linkedinStatus">
                        Not Checked
                    </strong>
                </div>

                <div class="validation-item">
                    <span>GitHub</span>
                    <strong id="githubStatus">
                        Not Checked
                    </strong>
                </div>

            </div>

            <!-- KEYWORDS -->

            <div class="keywords">

                <h2>🔑 Detected Keywords</h2>

                <p class="keyword-list" id="keywords">
                    No keywords detected yet.
                </p>

            </div>

        </div>

    </div>

    <div class="footer">
        JavaScript Lab Project • String Processing •
        Regular Expressions • Client-Side Validation
    </div>

</div>

<script>

    function generatePortfolio() {

        /* GET INPUT VALUES */

        let name =
            document.getElementById("name").value.trim();

        let email =
            document.getElementById("email").value.trim();

        let phone =
            document.getElementById("phone").value.trim();

        let skillsText =
            document.getElementById("skills").value.trim();

        let bio =
            document.getElementById("bio").value.trim();

        let linkedin =
            document.getElementById("linkedin").value.trim();

        let github =
            document.getElementById("github").value.trim();

        /* STRING FUNCTIONS */
```

```
        // Convert name to uppercase
        let upperName =
            name.toUpperCase();

        // Get first character
        let firstLetter =
            name.charAt(0).toUpperCase();

        // Count characters
        let bioLength =
            bio.length;

        // Convert skills into an array
        let skills =
            skillsText
            .split(",")
            .map(skill => skill.trim())
            .filter(skill => skill !== "");

        // Convert array back into string
        let skillString =
            skills.join(", ");

        /* REGULAR EXPRESSIONS */

        let emailRegex =
            /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

        let phoneRegex =
            /^[6-9]\d{9}$/;

        let nameRegex =
            /^[A-Za-z ]{2,50}$/;

        let linkedinRegex =
            /^https?:\/\/(www\.)?linkedin\.com\/.+/i;

        let githubRegex =
            /^https?:\/\/(www\.)?github\.com\/.+/i;

        /* VALIDATION */

        checkStatus(
            "emailStatus",
            emailRegex.test(email)
        );

        checkStatus(
            "phoneStatus",
            phoneRegex.test(phone)
        );

        checkStatus(
            "nameStatus",
            nameRegex.test(name)
        );

        checkStatus(
            "linkedinStatus",
            linkedinRegex.test(linkedin)
        );

        checkStatus(
            "githubStatus",
            githubRegex.test(github)
        );

        /* PORTFOLIO PREVIEW */

        document.getElementById("previewName")
            .textContent =
            upperName || "Student Name";

        document.getElementById("previewEmail")
            .textContent =
            email || "student@example.com";
```

```
        document.getElementById("previewBio")
            .textContent =
            bio || "Your portfolio bio will appear here.";

        document.getElementById("avatar")
            .textContent =
            firstLetter || "S";

        /* DISPLAY SKILLS */

        let skillHTML = "";

        if (skills.length > 0) {

            skills.forEach(function(skill) {

                skillHTML +=
                    `<span class="skill">${skill}</span>`;

            });

        } else {

            skillHTML =
                `<span class="skill">Your Skills</span>`;
        }

        document.getElementById("previewSkills")
            .innerHTML = skillHTML;

        /* LINKS */

        document.getElementById("linkedinLink")
            .href = linkedin || "#";

        document.getElementById("githubLink")
            .href = github || "#";

        /* TEXT ANALYTICS */

        // Character count
        document.getElementById("charCount")
            .textContent = bioLength;

        // Word count
        let words =
            bio === ""
            ? []
            : bio.split(/\s+/);

        document.getElementById("wordCount")
            .textContent = words.length;

        // Skill count
        document.getElementById("skillCount")
            .textContent = skills.length;

        // Count vowels using Regex
        let vowels =
            bio.match(/[aeiou]/gi);

        document.getElementById("vowelCount")
            .textContent =
            vowels ? vowels.length : 0;

        /* KEYWORD DETECTION */

        let importantKeywords = [

            "javascript",
            "python",
            "java",
            "html",
            "css",
            "react",
            "sql",
            "ai",
```

```
            "machine learning",
            "data science",
            "developer"

        ];

        let lowerBio =
            bio.toLowerCase();

        let foundKeywords = [];

        importantKeywords.forEach(function(keyword) {

            if (lowerBio.includes(keyword)) {

                foundKeywords.push(keyword);

            }

        });

        if (foundKeywords.length > 0) {

            document.getElementById("keywords")
                .textContent =
                foundKeywords.join(" • ");

        } else {

            document.getElementById("keywords")
                .textContent =
                "No keywords detected yet.";

        }

    }

    /* VALIDATION FUNCTION */

    function checkStatus(elementId, isValid) {

        let element =
            document.getElementById(elementId);

        if (isValid) {

            element.textContent = "✓ Valid";
            element.className = "valid";

        } else {

            element.textContent = "✗ Invalid";
            element.className = "invalid";

        }

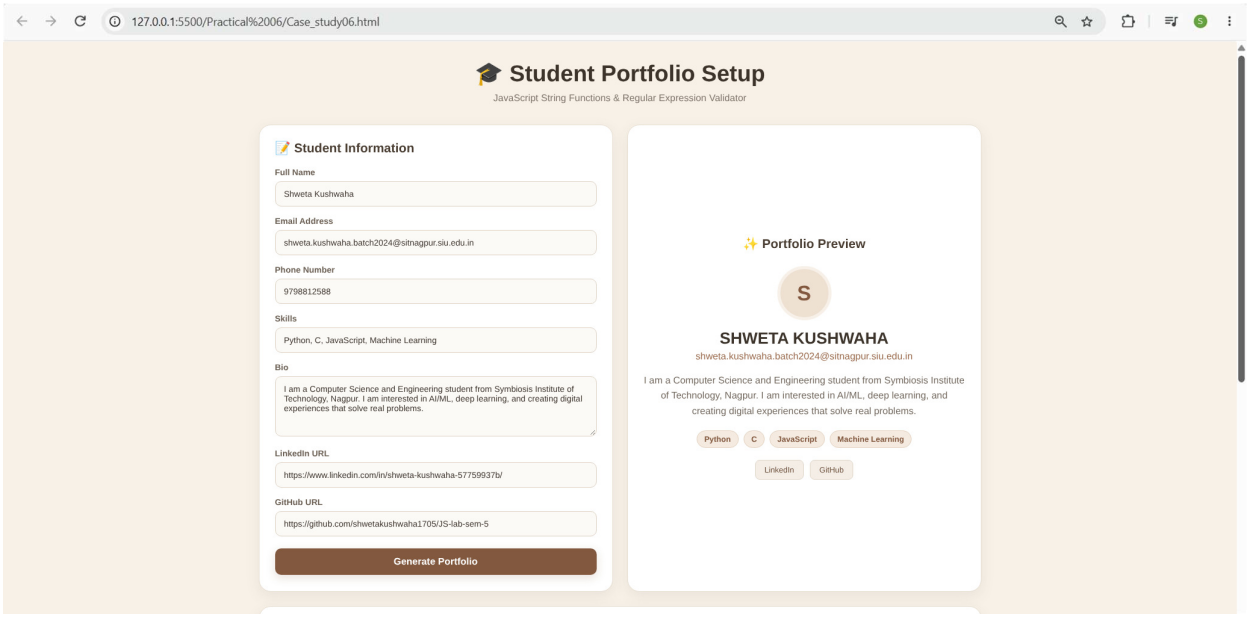
    }

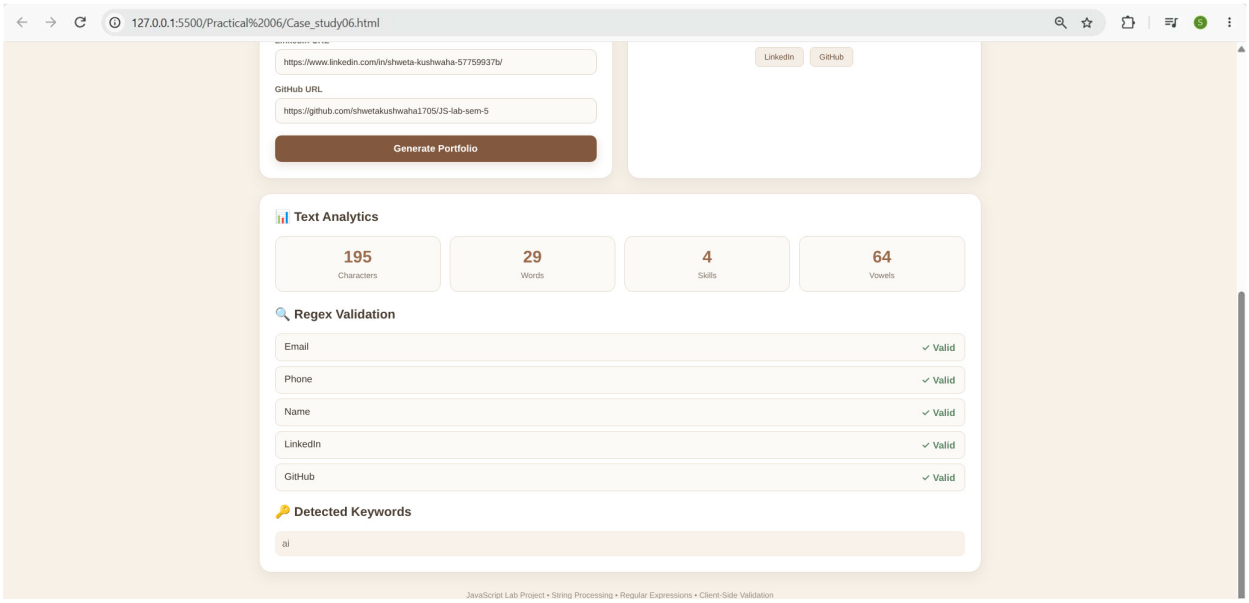
</script>

</body>
</html>
```

7. Output of the case study :

Name : Shweta Kushwaha PRN : 24070521001 File path :
http://127.0.0.1:5500/Practical%2002/Casestudy_06_Shweta_Kushwaha_1001.html





8. Conclusion

The Student Portfolio Setup application successfully demonstrates the implementation of JavaScript string functions and regular expressions for clientside input validation, text formatting, data extraction, and summary analysis. It effectively validates email addresses and phone numbers, processes user-entered text, generates useful analytics, and provides immediate feedback, making it a practical example of string processing and regex-based validation in web development.